ZCAS UNIVERSITY

MASTER OF SCIENCE IN INFORMATION TECHNOLOGY
MASTER OF SCIENCE IN COMPUTER SCIENCE

CAS 6181

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

FINAL EXAMINATION

TUESDAY 3rd JUNE 2025

TIME: 14.00 – 17.00 HOURS

TIME ALLOWED: THREE HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS

1. Section A: Question One in Section A is **compulsory**.
2. Sections B: Answer Two (2) questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.
5. Answers must reflect the Masters Degree level.

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SECTION A: QUESTION 1 IS COMPULSORY AND MUST BE ATTEMPTED. CALCULATIONS SHOWN EARN MARKS, OTHERWISE MARKS ARE LOST

Question 1:

Scenario: Read the scenario in Appendix A, (also sent to you earlier), and then answer the following questions:

a) In terms of Expert Systems, "Analyse how MediAI's conflict resolution strategy uses meta-rules when facing contradictory conclusions about drug prescriptions."

(10 Marks)

b) Resolve the following using Logic Agents:

"Convert the medical rule 'If fever >39°C AND rash present THEN consider viral infection' into propositional logic. Create a truth table and identify critical cases."

(10 Marks)

c. Considering Uncertainty Management, resolve the following:

"Given P(DiseaseA)=0.15, P(SymptomX|DiseaseA)=0.8, and P(SymptomX|¬DiseaseA)=0.05, calculate the posterior probability of DiseaseA using Bayesian reasoning when SymptomX is present." (10 Marks)

d. With Fuzzy Logic Application, "Design a fuzzy inference system for '**fever severity**' with three membership functions. Describe how MediAI uses this for pediatric cases." (10 Marks)

e) "Explain why A* was chosen over best-first search for MediAI's diagnostic pathfinding, considering the system's constraints." (10 Marks)

[TOTAL: 50 MARKS]

SECTION B: ATTEMPT TWO QUESTIONS ONLY FROM THIS SECTION

Question 2:

Case Study: SmartFarm AI – Optimizing Crop Yield Using Neural Networks and Genetic Algorithms

Background:

SmartFarm AI is a startup that aims to revolutionize precision agriculture using Artificial Intelligence and Machine Learning. The company has deployed IoT-enabled sensors across multiple farms to collect real-time data such as soil moisture, temperature, humidity, nutrient levels, sunlight exposure, and historical yield patterns. This data is used to predict optimal irrigation schedules, fertilizer needs, and planting strategies to maximize crop yield.

To handle the complex nonlinear relationships in environmental data and predict crop yield, SmartFarm AI has implemented **Artificial Neural Networks (ANNs)**. To further optimize the structure and hyperparameters of the neural networks (such as the number of hidden layers, neurons per layer, learning rate, and activation functions), SmartFarm AI incorporates a **Genetic Algorithm (GA)** as a meta-heuristic optimizer.

The company has reported increased yields of up to 20% in pilot farms. However, the team is now tasked with improving model generalization, speeding up convergence, and making the system adaptable to different crops and climate zones.

a) Describe the key components of a Genetic Algorithm. (15 Marks)

b) Explain the process of training a neural network using backpropagation. (10 Marks)

[TOTAL: 25 MARKS]

Question 3:

Case Study: Optimizing Urban Traffic Flow with Swarm Intelligence

Background:

The metropolitan city of NovaCity has faced increasing problems with traffic congestion, unpredictable flow, and inefficient signal timing at intersections. Traditional traffic light systems, which use static schedules, have failed to adapt to real-time conditions, leading to long delays and increased pollution levels.

In response, the city's Department of Transportation partnered with a tech startup to develop an **adaptive traffic management system** using **Swarm Intelligence (SI)** principles inspired by the behavior of ants and bees. The system, known as **SwarmTraffic**, leverages **Ant Colony Optimization (ACO)** and **Bee Algorithms** to optimize routing and signal coordination dynamically.

How It Works:

- **Ant Colony Optimization (ACO):** Vehicles are treated like ants, leaving behind digital pheromone trails based on speed, congestion, and travel time. Other vehicles read these trails to choose optimal paths in real time.
- **Bee Algorithm:** Scout agents explore alternate routes; based on efficiency, they communicate better paths to the rest of the vehicle agents, who then follow and reinforce successful routes.
- **Decentralized Agents:** Traffic lights act as autonomous agents, adjusting their timing based on swarm behavior rather than a central controller.

The system was deployed in a test zone for 6 months and led to:

- **30% reduction** in average travel time
- **20% reduction** in CO_2 emissions
- **Increased throughput** during rush hours

Questions:

a) Explain how Ant Colony Optimization (ACO) was used in the SwarmTraffic system. Discuss the role of the Bee Algorithm in the case study and how it differs from ACO. (20 Marks)

b) How does the concept of "stigmergy" apply to Swarm Intelligence in traffic systems? (5 Marks)

[TOTAL: 25 MARKS]

Question 4:

Case Study: Ethical Deployment of Autonomous Robots in Smart Warehousing

A multinational e-commerce company introduced **Autonomous Mobile Robots (AMRs)** at its Rotterdam warehouse to improve order fulfillment and inventory management. These robots navigate dynamic environments using **SLAM (Simultaneous Localization and Mapping)** and **AMCL (Adaptive Monte Carlo Localization)**, and rely on **LIDAR, depth cameras**, and **ultrasonic sensors** to detect obstacles and human workers in real-time.

To ensure **safe and ethical behaviour**, the robots are governed by **Isaac Asimov's Three Laws of Robotics**:

1. **A robot may not injure a human being or, through inaction, allow a human being to come to harm.**
2. **A robot must obey human orders unless doing so conflicts with the First Law.**
3. **A robot must protect its own existence unless this conflicts with the First or Second Law.**

These laws were embedded into the robots' **ethical decision-making engine**, guiding task execution, navigation, and conflict resolution. The robots operate on **ROS**

(Robot Operating System), enabling modular integration of perception, planning, and control systems.

Results after 18 months include:

- **Zero human-robot collisions**, owing to proactive safety protocols.
- **Increased productivity** through safe human-robot collaboration.
- Enhanced **trust and compliance** with AI ethics standards.

The project demonstrates how robotics, when aligned with ethical principles, can transform industrial operations while ensuring human safety and cooperation.

In Contrast: Military and Autonomous Weapons Systems

Outside the civilian domain, **autonomous weapons systems (AWS)**, or **"killer robots,"** are being developed for battlefield use. These systems may identify and engage targets **without direct human intervention**. Unlike AMRs, AWS are **not typically bound** by Asimov's laws and raise critical questions about:

- **Accountability** for unintended harm.
- **Moral and legal responsibility** in lethal decision-making.
- **Compliance with international humanitarian law (IHL)** and human rights.

Global debates continue on whether to **regulate or ban killer robots**, while nations invest in increasingly autonomous military systems.

Questions:

a) For a robot to have the best level of success, 4 R's were created to ensure the best out put for the robot. Explain the 4Rs. (10 Marks)

b) Discuss whether Asimov's Three Laws of Robotics can be applied to military robots? Justify your answer. (10 Marks)

c) Give the details of the components of Robots. (5 Marks)

[TOTAL: 25 MARKS]

APPENDIX A

Case Study: AI-Powered Diagnostic System for Rare Diseases

Overview

This case study examines MediAI, an intelligent diagnostic system deployed at Johns Hopkins Medical Center that combines machine learning with symbolic reasoning to identify rare genetic disorders. The system integrates heterogeneous data sources including genomic data, patient histories, and medical imaging.

Technical Implementation

1. Intelligent Agents Architecture

- Uses a hybrid agent architecture with:
  - Reactive components for real-time patient monitoring
  - Deliberative components for differential diagnosis
  - Utility-based decision making for treatment recommendations

2. Problem Solving Through Search

- Implements A* search algorithm for optimal diagnostic pathfinding
- Maintains branching factor < 3 through symptom clustering
- Heuristic function based on biomarker prevalence

3. Knowledge Representation

- Dual representation system:
  - Frame-based system for patient profiles
  - OWL ontology for disease knowledge base
  - Probabilistic graphical models for symptom relationships

4. Uncertainty Management

- Bayesian network with 152 nodes for symptom-disease mapping
- Dempster-Shafer theory for conflicting test results

- Monte Carlo simulation for treatment outcome prediction

5. Logic-Based Reasoning

- Propositional logic rules for drug interaction checks
- First-order logic for inheritance pattern analysis
- Fuzzy logic for symptom severity assessment

Challenges & Solutions

Challenge	AI Solution	Outcome
Data Sparsity	Generative adversarial networks for synthetic rare disease cases	38% improvement in detection rate
Expert Disagreement	Belief revision system with justification preservation	92% consensus achievement
Temporal Reasoning	Markov decision process for disease progression	27% earlier interventions

Results (2024 Implementation)

- Reduced diagnostic time from 6.2 to 1.8 months
- Increased accuracy from 68% to 89% compared to human experts
- Prevented 12,000+ adverse drug reactions through logic checks.

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